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1. INTRODUCTION

With the rapid advancement of online education, educational institutions require efficient systems to manage academic and administrative operations digitally. Many small and medium-sized academies still rely on spreadsheets and manual record-keeping systems, which often lead to redundancy, inconsistencies, and difficulties in maintaining accurate records.

The Online Academy Management System (OAMS) is a centralized relational database system developed to manage students, instructors, courses, enrollments, attendance, examinations, payments, and academic performance efficiently. The system ensures proper organization of academic data while maintaining data integrity and reducing redundancy.

The project is developed using relational database concepts and SQL implementation techniques learned in the Database Management Systems course.

2. PROBLEM STATEMENT

Traditional academic management methods face several limitations, including:

- Data redundancy
- Inconsistent record updates
- Difficulty in generating reports
- Limited analytical querying capability
- Poor management of attendance and fee records
- Lack of centralized academic information

There is a need for a structured and normalized database system that ensures efficient data storage, proper relationships among entities, and support for advanced reporting and querying.

3. OBJECTIVES

The objectives of the Online Academy Management System are:

- To design a normalized relational database
- To reduce redundancy and update anomalies
- To establish proper entity relationships using foreign keys
- To enforce data integrity through constraints
- To support advanced SQL queries and reporting
- To manage student enrollments and attendance efficiently
- To maintain accurate payment and examination records
- To generate analytical academic and financial reports

4. SCOPE OF THE SYSTEM

The system covers the following modules:

4.1 Student Management

- Student registration
- Student profile management
- Enrollment tracking

4.2 Instructor Management

- Instructor information management

- Course assignment handling

4.3 Course & Batch Management

- Course creation
- Batch scheduling
- Batch management

4.4 Enrollment Management

- Student course registration
- Enrollment status management

4.5 Attendance Management

- Attendance recording
- Attendance tracking

4.6 Payment Management

- Fee payment records
- Pending dues tracking
- Revenue reporting

4.7 Examination & Results

- Exam scheduling
- Marks management
- Result generation

4.8 Course Materials

- Learning resource management
- Batch-specific material uploads

5. FUNCTIONAL REQUIREMENTS

The system provides the following functionalities:

- Add, update, and delete student records
- Add and manage instructors
- Create and manage courses

- Create batches for courses
- Enroll students in batches
- Record attendance
- Record fee payments
- Conduct exams
- Store examination results
- Generate academic reports
- Generate payment summaries
- Identify students with low attendance
- Identify students with pending fees

6. NON-FUNCTIONAL REQUIREMENTS

The system satisfies the following non-functional requirements:

- Data consistency
- Data integrity
- Reduced redundancy
- Scalable database structure
- Efficient query execution
- Reliable data storage
- Secure relational structure
- Easy maintenance and extensibility

7. SYSTEM MODULES

The Online Academy Management System consists of the following modules:

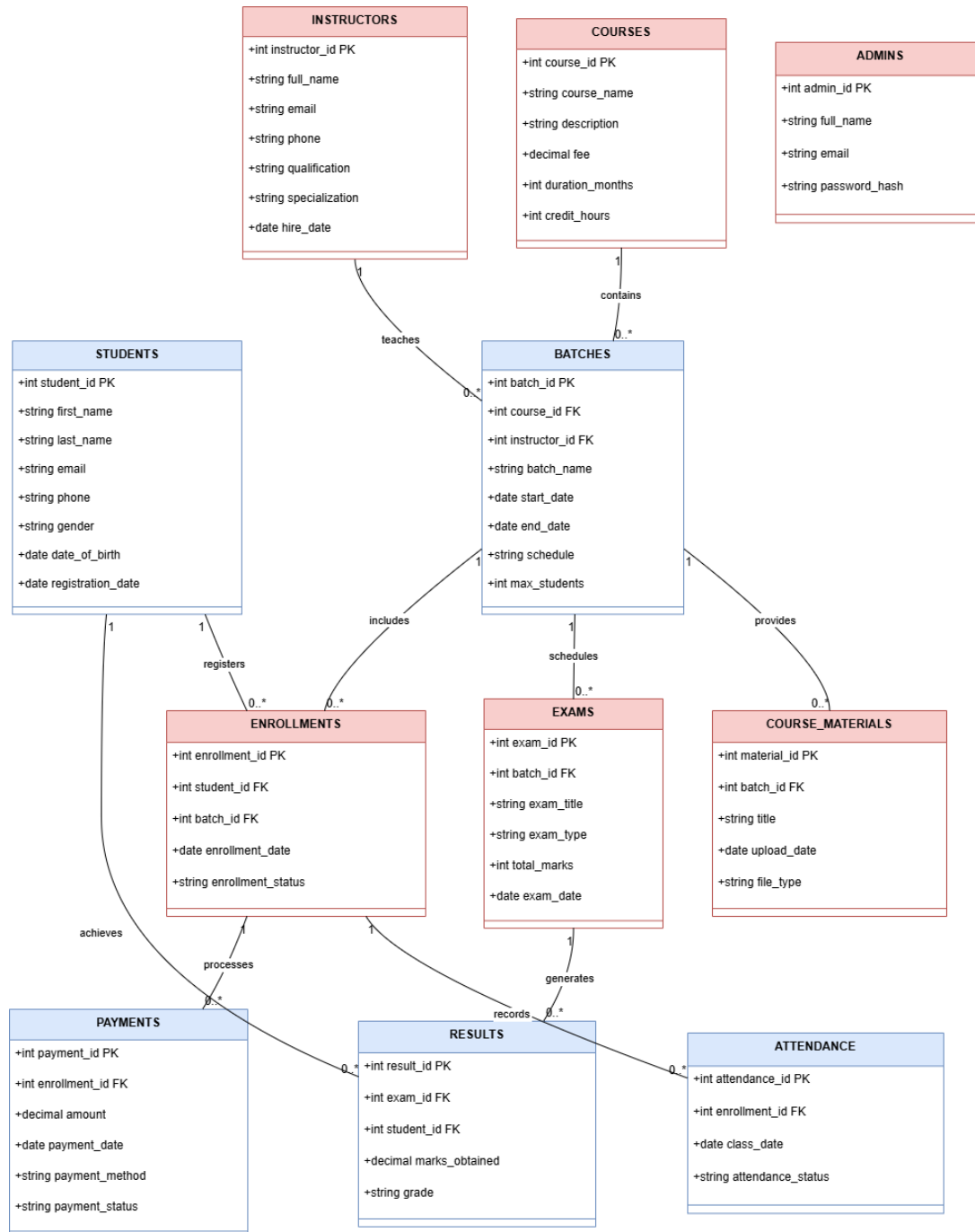
Module	Description
Students	Stores student information
Instructors	Stores instructor records
Courses	Stores course information
Batches	Manages course batches
Enrollments	Connects students with batches
Attendance	Maintains attendance records
Payments	Stores fee transactions
Exams	Manages exam details
Results	Stores student marks and grades
Course Materials	Stores learning resources
Admins	Stores administrator accounts

8. ENTITY RELATIONSHIP DIAGRAM (ERD)

The ERD represents the logical structure of the database and relationships between entities. Main relationships include:

- One Course can have many Batches
- One Instructor can teach many Batches
- One Student can enroll in many Batches
- One Batch can contain many Students
- One Enrollment can have many Attendance records

- One Enrollment can have many Payments
- One Batch can have many Exams
- One Exam can generate many Results
- One Student can have many Results



9. RELATIONAL SCHEMA

The relational schema of the system is as follows:

STUDENTS

(Student_ID PK, First_Name, Last_Name, Email, Phone, Gender, Date_Of_Birth, Registration_Date)

INSTRUCTORS

(Instructor_ID PK, Full_Name, Email, Phone, Qualification, Specialization, Hire_Date)

COURSES

(Course_ID PK, Course_Name, Description, Fee, Duration_Months, Credit_Hours)

ADMINS

(Admin_ID PK, Full_Name, Email, Password_Hash)

BATCHES

(Batch_ID PK, Course_ID FK, Instructor_ID FK, Batch_Name, Start_Date, End_Date, Schedule, Max_Students)

ENROLLMENTS

(Enrollment_ID PK, Student_ID FK, Batch_ID FK, Enrollment_Date, Enrollment_Status)

ATTENDANCE

(Attendance_ID PK, Enrollment_ID FK, Class_Date, Attendance_Status)

PAYMENTS

(Payment_ID PK, Enrollment_ID FK, Amount, Payment_Date, Payment_Method, Payment_Status)

EXAMS

(Exam_ID PK, Batch_ID FK, Exam_Title, Exam_Type, Total_Marks, Exam_Date)

RESULTS

(Result_ID PK, Exam_ID FK, Student_ID FK, Marks_Obtained, Grade)

COURSE_MATERIALS

(Material_ID PK, Batch_ID FK, Title, Upload_Date, File_Type)

10. DATABASE NORMALIZATION

The database design follows normalization principles to eliminate redundancy and improve consistency.

First Normal Form (1NF)

- All attributes contain atomic values

- No repeating groups exist

Second Normal Form (2NF)

- All non-key attributes are fully dependent on the primary key
- Partial dependency is removed

Third Normal Form (3NF)

- Transitive dependencies are eliminated
- Non-key attributes depend only on the primary key

Normalization improves:

- Data consistency
- Query performance
- Storage efficiency
- Maintainability

11. SQL SCHEMA IMPLEMENTATION

The database schema was implemented using MySQL.

The implementation includes:

- CREATE TABLE statements
- Primary Keys
- Foreign Keys
- CHECK constraints
- UNIQUE constraints
- AUTO_INCREMENT attributes

The schema ensures referential integrity and proper relational mapping between entities.

Example:

Students with Pending Payments

```
CREATE TABLE students (  
    student_id INT PRIMARY KEY AUTO_INCREMENT,  
    first_name VARCHAR(50) NOT NULL,
```



```
last_name VARCHAR(50) NOT NULL,  
email VARCHAR(100) UNIQUE NOT  
NULL  
);
```

Attendance Percentage Report

```
SELECT  
    student_id,  
    COUNT(attendance_id  
    )  
FROM attendance  
GROUP BY student_id;
```

12. BENEFITS OF THE SYSTEM

The Online Academy Management System provides the following benefits:

- Centralized academic management
- Improved data organization
- Reduced redundancy
- Efficient reporting
- Faster record retrieval
- Better attendance tracking
- Improved fee management
- Better academic performance monitoring

13. FUTURE ENHANCEMENTS

Future improvements may include:

- Web-based application integration
- Mobile application support
- Student portal
- Online examination system
- AI-based student performance prediction

- Notification and email integration
- Automated attendance system
- Learning Management System (LMS) integration
- Cloud database deployment

14. **CONCLUSION**

The Online Academy Management System successfully demonstrates the implementation of relational database concepts using SQL and normalization techniques.

The project effectively manages academic operations including student enrollment, attendance tracking, fee management, examinations, and result processing through a structured and normalized database design.

The system ensures data consistency, minimizes redundancy, and supports advanced analytical queries for academic and administrative reporting. The project also provides a scalable foundation for future enhancements and real-world deployment.

The developed database system fulfills the objectives of the project and demonstrates practical application of Database Management Systems concepts in a real-world educational environment.